ok, now the final idea has been given to the "Virtual Labs" pdf and the guidelines in the "Initial Brief" pdf and the "ILP-1.pdf" says about our work that we did based on the kind of companies we are targeting and a basic analysis of them.

We want to concentrate at one particular tab of the project first and we want to come up with KPI that a person from this kind of industry will understand will feel as if an savior of his problems. Lets get it right for one tab and then we proceed with other tabs. When we are satisfactorily done for one particular industry with all the necessary tabs, we will make slight tweaks so that we can customize the project to industry specifics metrics/values so that the potential customer will feel a greater feel for it.

so for now, lets get the kpi the and approximate values that should be shown in the interface for a really good demo play around.

Using the below dataset and trying to find the values for the below questions so that we have approximate values in the dashboard (tab 1) and the valid sources linked to these numbers.

Dataset reference: FMCG Multi-Country Sales Dataset

1. Which SKUs generate the highest share of revenue, gross margin, and unit sales?
2. What percentage of total revenue is contributed by the top 10%, 20%, and 30% of SKUs?
3. Which SKUs contribute less than 1% of total revenue but still consume operational resources?
4. Which SKUs have low revenue contribution but high lead times, stockouts, or promo dependency?
5. Which SKUs are margin-dilutive despite high sales volume?
6. Which SKUs have the lowest gross margin percentage across categories?
7. Which categories contain the highest concentration of low-performing SKUs?
8. Which SKUs are most dependent on promotions for sales uplift?
9. Which SKUs experience margin erosion during promotional periods?
10. Which SKUs show unstable demand patterns based on volatility_class?
11. Which SKUs have the highest stockout frequency?
12. Which SKUs require disproportionately high safety stock compared to their revenue contribution?
13. Which suppliers are associated with the highest operational complexity?
14. How many suppliers exist per category and per SKU cluster?
15. Which supplier relationships contribute very little business value?
16. Which countries or regions contribute high SKU complexity but low profitability?
17. Which channels (Modern Trade, Traditional Trade, Wholesale, E-commerce) generate the highest margin efficiency?
18. Which channels show the highest demand volatility and stockout risk?
19. Which SKUs have the longest lead times and highest forecasting instability?
20. Which SKUs show strong seasonality patterns that affect forecast reliability?
21. Can we segment SKUs into:

- Keep
- Consolidate
- Rationalize
- Grow

based on commercial value vs operational complexity?

22. Build a Commercial Value Score using:
    - Revenue
    - Gross Margin
    - Revenue Growth
    - Demand Stability
23. Build an Operational Complexity Score using:
    - Lead Time
    - Supplier Dependency
    - Promo Dependency
    - Stockout Frequency
    - Volatility Class
24. Which SKUs fall into:
    - High Value / Low Complexity
    - High Value / High Complexity
    - Low Value / High Complexity
    - Low Value / Low Complexity
25. Estimate a Portfolio Complexity Index for the enterprise using:
    - SKU proliferation
    - Supplier fragmentation
    - Volatility
    - Low velocity SKUs
    - Lead time instability
    - Promo dependency
26. Estimate a Cannibalization Risk Score by checking whether promotions or spikes in one SKU reduce sales of similar SKUs in the same category.
27. Which SKUs have the highest operational burden relative to revenue contribution?
28. Which SKUs should be prioritized for rationalization with minimal revenue impact?
29. Simulate the impact of eliminating the bottom 10%, 20%, and 30% of low-performing SKUs.
30. For each rationalization simulation, calculate projected changes in:
    - Gross Margin
    - Supplier Count
    - Forecast Accuracy
    - Safety Stock
    - Stockout Frequency
    - Operational Complexity
    - Inventory Carrying Cost
31. Identify the top 10 portfolio risks currently affecting profitability and operational efficiency.

32. Identify the top 10 AI-generated rationalization opportunities with estimated business impact.
33. Generate executive-level narrative insights such as:
    - “31% of SKUs contribute only 4% of revenue but consume 22% of supplier relationships.”
    - “Top 12% of SKUs drive 68% of portfolio margin.”
    - “Removing 240 low-productivity SKUs could improve margin by 2.4% while reducing operational complexity by 14%.”
34. Highlight which KPIs are directly supported by dataset columns and which are estimated/model-derived.
35. Clearly identify assumptions used for:
    - Cannibalization estimation
    - Complexity scoring
    - Rationalization impact simulation
    - Forecast risk modeling

Data extraction from the dataset present in this notebook : PPL_demo_data.ipynb

Analysis done and presented : [Colab_Notebook_interpreted]

It has not been properly documented and was feed to llm models for extraction of questions and answers from the notebook. The answers for the above questions are given below.

FMCG SKU Performance Analysis Report

Portfolio Data Collection and Insight Documentation

Executive Highlights

- **The Revenue Concentration:** The top **10% of SKUs** drive **27.81%** of total revenue, while the top **30%** generate **62.88%**. Conversely, **66.7% of SKUs** contribute **<1%** of revenue each, yet drive 100% of current supplier administrative overhead.
- **Rationalization Impact:** Removing all 35 "Rationalize" SKUs cuts safety stock by **42.2%** ($246\text{M} \rightarrow 142\text{M}$), but introduces a **27.08% revenue tail risk**.
- **Operational Friction: Dairy** holds the highest concentration of low-performing items (**27.78%**).

Detailed Findings & Extracted Responses

Q1. Top SKUs by Revenue, Gross Margin, and Unit Sales

- **By Net Sales:** BrandF Water (17.03M), BrandC Chips (16.00M), BrandB Chips (13.03M), BrandC Toothpaste (12.97M), BrandB Soap (12.50M)

- **By Gross Margin:** BrandF Water (6.83M), BrandC Chips (5.80M), BrandB Chips (5.19M), BrandB Soap (5.00M), BrandD Cheese (4.96M)
- **By Unit Sales:** BrandA Soda (1,792,381), BrandF Soda (1,662,175), BrandE Chocolate (1,455,704), BrandF Soap (1,286,230), BrandD Chips (1,253,987)

Q2. Revenue Contribution by Top SKU Tiers

- **Top 10% of SKUs** (10 SKUs) $\rightarrow$ **27.81%** of total revenue
- **Top 20% of SKUs** (20 SKUs) $\rightarrow$ **48.51%** of total revenue
- **Top 30% of SKUs** (30 SKUs) $\rightarrow$ **62.88%** of total revenue

Q3. Low-Revenue Resource Consumers (<1% Revenue)

68 SKUs generate less than 1% revenue share but still heavily drain operational resources through stockouts or high promotional inclusion.

- *Sample SKUs:* BrandB Yogurt, BrandA Softener, BrandF Biscuits, BrandE Chips, BrandC Softener, BrandF Chips, BrandA Milk, BrandC Soda, BrandE Cheese, BrandA Soap.

Q4. Low Revenue + High Operational Friction Profile

21 SKUs fall into the bottom 25% of revenue while causing peak operational friction.

- *Sample Highlights:* * **BrandA Softener:** Lead time 6.51 days, 293 stockouts
    - **BrandE Chips:** Lead time 6.47 days, 295 stockouts, high promotion dependency
    - *Other notable items:* BrandC Softener, BrandF Chips, BrandA Milk, BrandC Soda, BrandE Cheese, BrandA Soap, BrandB Juice, BrandB Shampoo.

Q5. Margin-Dilutive SKUs Despite High Sales Volume

The average portfolio gross margin sits at **38.53%**. **12 high-volume SKUs** pull this average down significantly.

- *Top Dilutive Volumetric Items:* BrandA Soda (36.08%), BrandF Soda (36.27%), BrandE Water (35.87%), BrandC Toothpaste (35.84%), BrandC Chips (36.23%), BrandD Chocolate (36.15%), BrandA Water (36.01%), BrandD Nuts (35.91%), BrandE Soda (36.05%), BrandF Toothpaste (36.26%).

Q6. Lowest Gross Margin % Across Categories

The 10 SKUs with the absolute lowest margin profiles:

- BrandA Chocolate (35.83%), BrandB Softener (35.84%), BrandC Toothpaste (35.84%), BrandE Water (35.87%), BrandD Juice (35.90%), BrandD Nuts (35.91%), BrandA Energy Drink (35.91%), BrandD Water (35.92%), BrandB Milk (35.96%), BrandE Toothpaste (35.96%).

Q7. Categories with Highest Concentration of Low-Performing SKUs

Ranked by the percentage of category SKUs landing in the bottom 20% of revenue:

1. **Dairy** – 27.78%
2. **Beverages** – 20.83%
3. **Snacks** – 20.83%
4. **Home Care** – 16.67%
5. **Personal Care** – 16.67%

Q8. SKUs Most Dependent on Promotions

Top 10 SKUs ranked by the percentage of their total revenue generated during promotional periods:

- BrandD Water (27.97%), BrandD Juice (27.79%), BrandF Toothpaste (27.69%), BrandE Cheese (27.68%), BrandE Juice (27.62%), BrandA Shampoo (27.62%), BrandC Cheese (27.61%), BrandF Juice (27.59%), BrandB Chocolate (27.59%), BrandC Toothpaste (27.59%).

Q9. SKUs with Severe Margin Erosion During Promotions

Top 10 SKUs sorted by highest promotional margin erosion score:

- BrandC Toothpaste (15.49), BrandA Cleaner (15.13), BrandE Toothpaste (15.01), BrandB Milk (14.96), BrandC Nuts (14.88), BrandE Softener (14.87), BrandA Water (14.81), BrandA Soap (14.80), BrandF Juice (14.80), BrandE Biscuits (14.77).

Q10. SKUs with Unstable Demand (Volatility Class)

- **Stable:** 80 SKUs | **Variable:** 22 SKUs | **Unstable:** 0 SKUs

Note: No SKUs crossed the Coefficient of Variation threshold ($CV > 0.5$) to be classified as "Unstable". The corresponding section in the notebook output was blank.

Q11. Highest Stockout Frequency

Top 10 SKUs by total `stock_out_flag` occurrences:

- BrandC Biscuits (440), BrandF Soap (440), BrandB Energy Drink (427), BrandB Milk (417), BrandD Chips (416), BrandE Cheese (405), BrandA Soda (397), BrandA Toothpaste (394), BrandF Soda (393), BrandA Chips (392).

Q12. Disproportionately High Safety Stock vs. Revenue

Top 10 SKUs displaying the highest safety-stock-to-revenue ratios:

- BrandE Cheese (Ratio: 0.000248 | Net Sales: 847K)

- BrandF Soda (Ratio: 0.000222 | Net Sales: 2.02M)
- BrandA Biscuits (Ratio: 0.000209 | Net Sales: 1.10M)
- BrandE Water (Ratio: 0.000207 | Net Sales: 2.23M)
- *Remaining Top 10:* BrandA Shampoo (0.000204), BrandD Energy Drink (0.000194), BrandE Chips (0.000173), BrandC Yogurt (0.000169), BrandB Yogurt (0.000161), BrandF Chips (0.000144).

Q13. Suppliers with Highest Operational Complexity

The top 10 suppliers are uniformly structured, handling **102 SKUs each**:

- **Suppliers:** S001, S002, S003, S004, S005, S006, S007, S008, S009, S010
- *Financial Scope:* Net sales range between ~7.84M–7.96M with gross margins around ~3.02M–3.08M each.

Q14. Number of Suppliers per Category

The supply base is completely uniform. Each category interacts with exactly **60 unique suppliers**:

- Beverages: 60 | Dairy: 60 | Home Care: 60 | Personal Care: 60 | Snacks: 60

Q15. Supplier Relationships with Lowest Business Value

Bottom 10 suppliers ranked by total gross margin contribution (all managing 102 SKUs each):

- S058 (2.99M), S050 (2.99M), S022 (2.99M), S054 (2.99M), S052 (3.00M), S038 (3.00M), S021 (3.00M), S049 (3.01M), S018 (3.01M), S035 (3.01M).

Geographic & Channel Deep Dives

Q16. Regional SKU Complexity vs. Profitability

Country	SKU Count	Net Sales	Gross Margin %
Italy	102	137.2M	38.53%
Spain	100	106.7M	38.60%

Germany	98	88.5M	38.48%
France	80	42.6M	38.55%
Austria	80	43.0M	38.64%
Poland	80	42.4M	38.36%
Netherlands	45	12.5M	38.20%

Strategic Takeaway: The **Netherlands** presents an immediate assortment optimization opportunity: it holds the smallest footprint (45 SKUs) yet yields the lowest margin (38.20%). Italy and Spain drive volume but introduce massive complexity.

Q17 & Q18. Channel Performance, Volatility & Stockout Risks

Channel	Margin Efficiency	Volatility (CV)	Stockout Count
E-commerce	**38.56%** *(Highest)*	**0.061** *(Most Stable)*	7,907
Supermarket	38.53%	0.065	7,818
Hypermarket	38.52%	0.063	**15,907** *(Highest Risk)*

Convenience	**38.20%** *(Lowest)*	**0.069** *(Most Volatile)*	1,482

Operational Friction Matrices

Q19. Supply Chain Latency & Forecast Instability

- **Longest Lead Times (Days):** BrandA Nuts (6.54), BrandA Juice (6.54), BrandB Cleaner (6.53), BrandA Milk (6.53), BrandC Detergent (6.53), BrandD Soda (6.53), BrandA Soda (6.53), BrandD Toothpaste (6.53), BrandE Yogurt (6.52), BrandB Detergent (6.52).
- **Highest Forecasting Instability (CV):** BrandD Chocolate (0.242), BrandA Chocolate (0.237), BrandE Chocolate (0.232), BrandB Chocolate (0.227), BrandC Chocolate (0.224), BrandD Water (0.224), BrandD Juice (0.219), BrandE Water (0.217), BrandF Chocolate (0.216), BrandE Soda (0.215).

Q20. SKUs with Strongest Demand Seasonality

Ranked by monthly variance index; **Chocolate products overwhelmingly dominate portfolio seasonality**:

- BrandD Chocolate (0.239), BrandE Chocolate (0.238), BrandA Chocolate (0.236), BrandC Chocolate (0.229), BrandB Chocolate (0.229), BrandD Water (0.217), BrandF Chocolate (0.216), BrandD Juice (0.211), BrandF Energy Drink (0.206), BrandE Juice (0.206).

Strategic Portfolio Segmentation

Q21. Value vs. Complexity Matrix

Segmentation Thresholds applied at medians of normalized 0-1 metrics.

Segment	SKU Count	Description	Core Representative SKUs
Keep	35	High Value / Low Complexity. Core profit drivers.	BrandD Cheese, BrandD Toothpaste, BrandA Cheese, BrandB Soap

Grow	16	High Value / High Complexity. High revenue but heavy operational friction.	BrandC Chips, BrandC Toothpaste
Rationalize	35	Low Value / High Complexity. Primary SKU deletion candidates.	BrandF Soda, BrandA Chocolate, BrandD Chocolate
Consolidate	16	Low Value / Low Complexity. Stable, low impact; manage minimally.	*Generic low-impact baseline portfolio items*

Q22 & Q23. Scoring Methodology Formulas

$$\text{Commercial Value Score} = \text{Normalized}\big(\text{Net Sales} + \text{Gross Margin} + \text{YoY Growth (22}\rightarrow\text{23)} + (1 - \text{Volatility CV})\big)$$
$$\text{Operational Complexity Score} = \text{Normalized}\big(\text{Lead Time} + \text{Supplier Count Per SKU} + \text{Promo Dependency} + \text{Stockout Freq} + \text{Volatility Class Weight}\big)$$
Volatility Weights: Stable = 0.3, Variable = 0.6, Unstable = 1.0

Q25. Enterprise Portfolio Complexity Index (PCI)

- **Total Enterprise PCI = 0.5509**
- *Underlying Drivers:* Supplier Fragmentation Index (**1.2000**), SKU Proliferation Index (**1.0200**), Low Velocity SKU Concentration (**0.6667**), Lead Time Instability (**0.2014**), Average Portfolio Volatility CV (**0.1071**), Promo Dependency (**0.1100**).

Q26. Cannibalization Risk Assessment

- **Highest Risk Category: Beverages**. Heavy negative correlations were flagged between internal promo schedules and neighboring SKU unit sales, showing clear promotional "sales stealing".

Q27. Highest Operational Burden Relative to Revenue

Sorted by `op_burden_ratio` (Complexity Score / Commercial Value Score). All items fall within the **Rationalize** matrix segment:

- BrandF Soda (3.58), BrandA Chocolate (3.08), BrandD Chocolate (2.98), BrandD Water (2.61), BrandE Water (2.36), BrandE Juice (2.33), BrandD Juice (2.05), BrandC Water (1.94), BrandB Juice (1.87), BrandF Juice (1.86).

Rationalization Simulation Results

Q28, Q29 & Q30. Tiered SKU Deletion Projections

Simulation Scenario	SKUs Removed	Revenue Impact	Margin Impact	Safety Stock Freed	Supplier Reduction
Bottom 10% Tier	10	−3.58%	−3.46%	**+8.81%**	0.0%
Bottom 20% Tier	20	−9.00%	−8.81%	**+22.31%**	0.0%
Bottom 30% Tier	30	−14.05%	−13.82%	**+29.57%**	0.0%
Full "Rationalize" Segment	43	−27.08%	*N/A*	**+42.20%**	0.0%

Critical Simulation Constraint: Supplier base reduction remains at **0% across all models**. This reveals that all 60 suppliers are entirely interconnected across the portfolio; removing individual items fails to break supplier overhead dependencies.

Strategic Action Items

Q31 & Q32. Top Strategic Opportunities

1. **Execute Tier 1 Deletions:** Instantly remove the bottom 10% of SKUs to reclaim **8.8% of safety stock** capital while risking only 3.58% of top-line revenue.
2. **Restructure Supplier Networks:** Transition from 60 universal suppliers to specialized category clusters to deflate the high Supplier Fragmentation metric (1.20).

3. **Correct Promotional Depth:** Re-engineer promo mechanics for high-erosion lines like *BrandC Toothpaste* and *BrandA Cleaner* to salvage up to **15.5% in margin losses**.
4. **Address Category Drag:** Prune the Dairy category to address its high 27.78% concentration of low-performing SKUs.
5. **Realign Regional Assortments:** Adjust pricing and item mix in the Netherlands to bridge the gross margin gap against the Austrian benchmark (38.64%).

Q34 & Q35. Data Grounding & Structural Assumptions

- **Margin Determinations:** Derived using `net_sales` and `purchase_cost`, utilizing mean purchase cost data as a proxy for raw margins.
- **Rationalization Behavior:** Models assume revenue from a removed SKU is permanently lost to the business, rather than transferring cleanly to substitute active SKUs.
- **Forecast Risk Metric:** Leverages the Coefficient of Variation (CV) of monthly unit sales as the primary risk benchmark (CV < 0.2 = Stable, 0.2–0.5 = Variable).